

Abstract Linear Algebra

Prof. Lanier

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Pick up a handout. Try those warm-up problems.

Next up:

- Midterm revisions due on Friday.
- This week: Last thoughts on Chapter 4; Chapter 5
- Problem swap 2 problems due tonight.
- Quiz 9 on Friday (eigenvalues and eigenvectors.)
- HW₇ assigned on Friday. Last homework, half-sized.
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Facts about the
general 2×2 matrix A

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$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} \Rightarrow \lambda = \frac{\text{tr}(A)}{2} \pm \sqrt{\left(\frac{\text{tr}(A)}{2}\right)^2 - \det(A)}$$

Ch 5

inner products and orthogonality



These pairs of vectors are indistinguishable in $\mathbb{R}^2$,
with just its bare vector space structure.

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But by layering on additional structure—an inner product—
we can measure angles between vectors.

e.g. the dot product

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We can also study linear transformations
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Reminder:
going forward we are
only thinking about vector
spaces over $\mathbb{R}$ and $\mathbb{C}$.



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Zoom way out: norms and inner products

Important for rest of the course: Euclidean norm and dot product

Specific application: orthogonality, orthonormal bases

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For some operators,
we can find a similar operator
that is diagonal

$$\begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

diagonalization

Ch 6

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diagonalization

For ALL complex operators,
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$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \sim \begin{pmatrix} k & l & m \\ 0 & n & o \\ 0 & 0 & p \end{pmatrix}$$

Schur decomposition

Ch 9

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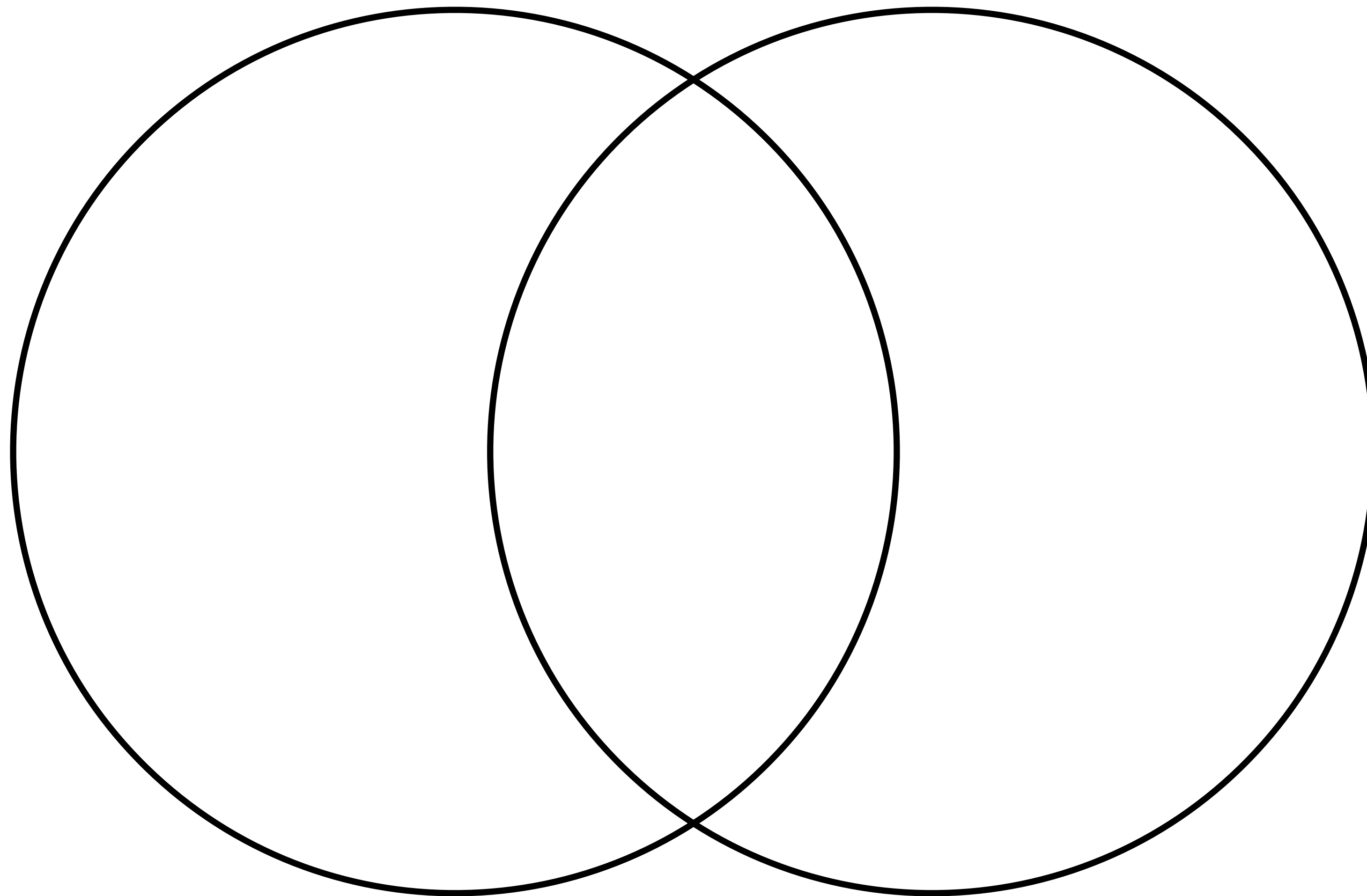
diagonalization

For ALL complex operators,
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of a particularly nice form

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \sim \begin{pmatrix} k & l & m \\ 0 & n & o \\ 0 & 0 & p \end{pmatrix} \sim \begin{pmatrix} k & 0 & 0 \\ 0 & n & 1 \\ 0 & 0 & p \end{pmatrix}$$

Schur decomposition Jordan canonical form

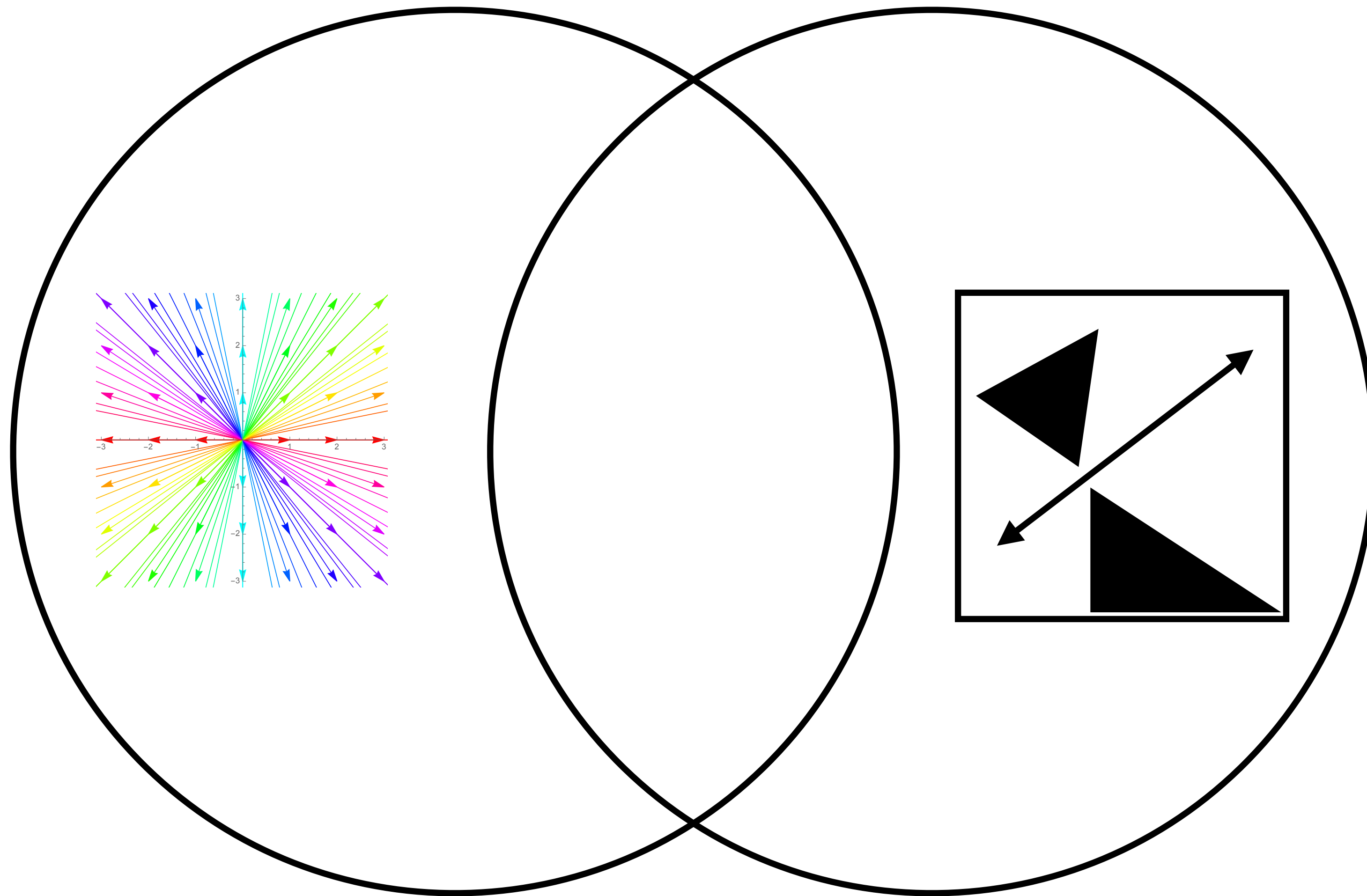


vector space

elements can add
and scale

metric space

there is a distance
function between
elements

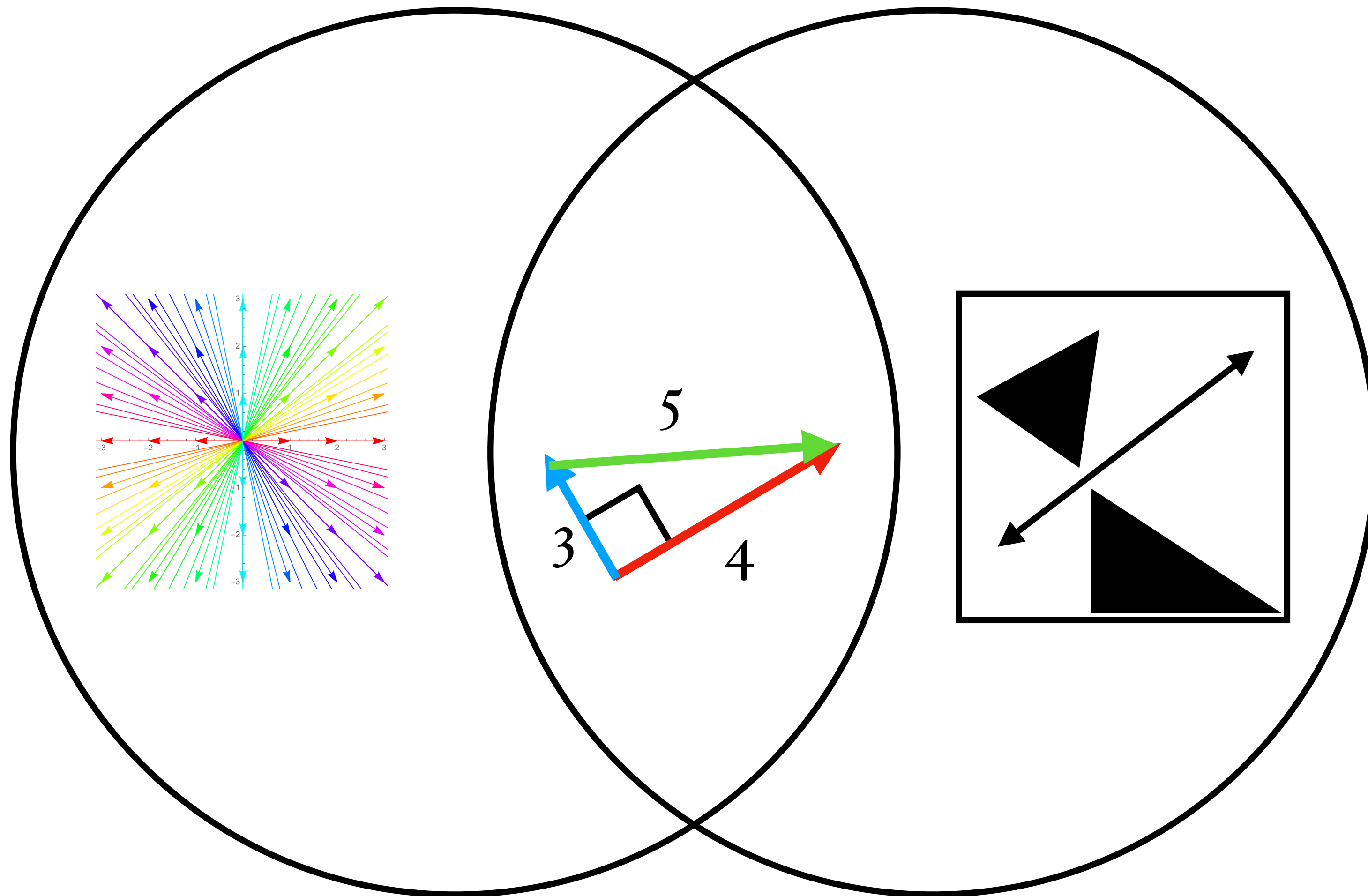


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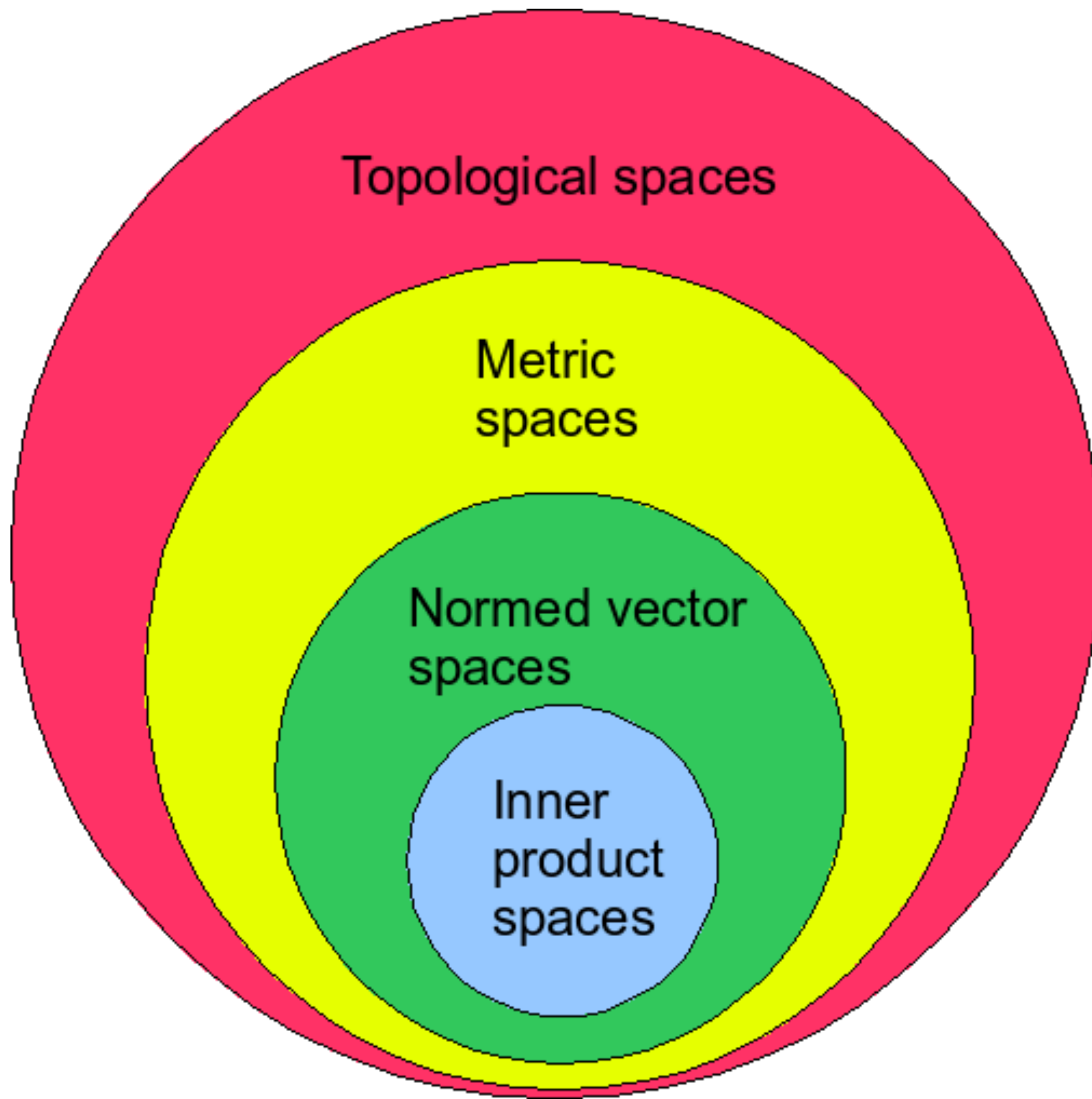


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Wikipedia, "Normed vector space"

Examples of norms (length functions) on $\mathbb{R}^n$ (using $n = 3$):

The usual Euclidean norm: $\begin{pmatrix} a \\ b \\ c \end{pmatrix} \rightarrow \sqrt{a^2 + b^2 + c^2}$

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The max norm: $\begin{pmatrix} a \\ b \\ c \end{pmatrix} \rightarrow \max(|a|, |b|, |c|)$

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The max norm: $\begin{pmatrix} a \\ b \\ c \end{pmatrix} \rightarrow \max(|a|, |b|, |c|)$ L^∞

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Defining properties of a norm on a vector space V .

1. Homogeneity: $\|\alpha \mathbf{v}\| = |\alpha| \cdot \|\mathbf{v}\|$ for all vectors $\mathbf{v}$ and all scalars α .
2. Triangle inequality: $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.
3. Non-negativity: $\|\mathbf{v}\| \geq 0$ for all vectors $\mathbf{v}$.
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A multilinear form on a vector space F^n is a function on k -tuples of vectors (for some fixed k) to the field F that is linear in each input/argument.

Examples:

The determinant

The dot product (a symmetric bilinear form)

The dot product of u and v is just $u^T v$ with usual matrix multiplication. It is useful to consider the result as a field element, rather than as a matrix. (standard inner product)

$$u \cdot v = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$$

$$u \cdot v = \|u\| \|v\| \cos \theta \quad (\text{in } \mathbb{R}^n)$$

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